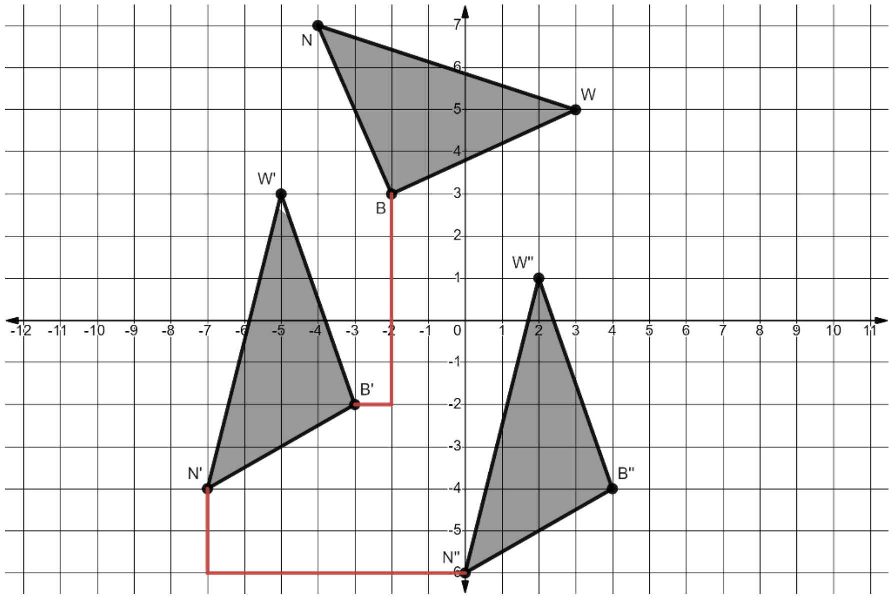
Geometry
Unit 4
Lesson 3 Practice

Name_Answer Key________
Date____________________
Period__________________

Triangles $BWN$, $B'W'N'$, and $B''W''N''$ are shown on the coordinate grid where $\triangle BWN$ is the preimage.




1. Describe the feature of $\triangle B'W'N'$ that indicates the rigid motion that was applied to $\triangle BWN$ that resulted in $\triangle B'W'N'$.
   Rotation $90^\circ$ counterclockwise about the origin.

2. Show the feature described by mapping the transformation of one vertex of $\triangle BWN$ to $\triangle B'W'N'$.

3. Complete the table.

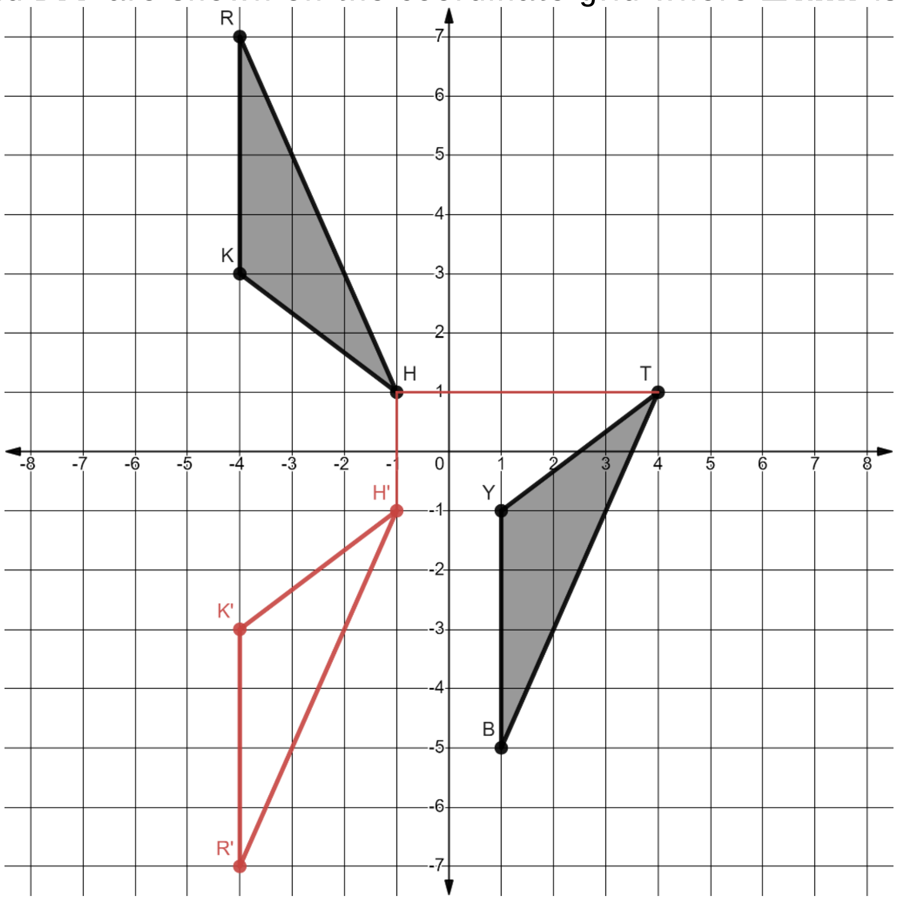
| Transformation | Mapping | Algebraic Description |
|---|---|---|
| Rotation $90^\circ$ counterclockwise about the origin. | $\triangle BWN \rightarrow \triangle B'W'N'$ | $(x, y) \rightarrow (-y, x)$ |

4. Map the transformation of one vertex of $\triangle B'W'N'$ to $\triangle B''W''N''$.

5. Complete the table.

| Transformation | Mapping | Algebraic Description |
|---|---|---|
| Translation right 7, down 2 | $\triangle B'W'N' \rightarrow \triangle B''W''N''$ | $(x, y) \rightarrow (x + 7, y - 2)$ |

Triangles $RKH$ and $BYT$ are shown on the coordinate grid where $\triangle RKH$ is the preimage.




6. Describe the feature of $\triangle TYB$ that indicates that more than one rigid motion was applied to $\triangle RKH$.
   The orientation and location changed.

7. Draw $\triangle R'K'H'$, the result of the first transformation that was applied to $\triangle RKH$.

8. Map the transformation of one vertex of $\triangle R'K'H'$ to $\triangle BYT$.

9. Complete the table.

| Transformation | Mapping | Algebraic Description |
|---|---|---|
| Reflect over $x$-axis | $\triangle RKH \rightarrow \triangle R'K'H'$ | $(x, y) \rightarrow (x, -y)$ |
| Translate right 5 up 2 | $\triangle R'K'H' \rightarrow \triangle BYT$ | $(x, y) \rightarrow (x + 5, y + 2)$ |

10. Explain how to know if the sequence of transformations preserves distance.
    The distance between each vertex in the image is the same.